Unit-2

Linear differential Equations of Higher order with constant Co-efficient

① $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0$ $\quad y(0) = 0,\ y'(0) = 15.$

Sol:- Given differential Equation,

$$\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0$$

In operator form,

$$(D^2 + 5D + 6)y = 0$$

$$f(D)y = 0$$

Auxillary Equation $f(m)y = 0$

$$m^2 + 5m + 6 = 0$$

$$m^2 + 3m + 2m + 6 = 0$$

$$m(m+3) + 2(m+3) = 0$$

$$(m+3)(m+2) = 0$$

$m + 3 = 0$ | $m + 2 = 0$

$m = -3$ | $m = -2$

$$m = -2, -3$$

General Solution is $y(x) = C_1 e^{-2x} + C_2 e^{-3x}$

$$y(x) = C_1 e^{-2x} + C_2 e^{-3x} \quad \textcircled{e}$$

where, C_1 and C_2 are arbitrary constants.

$\because y(0) = 0 \Rightarrow x = 0, y = 0$

$$0 = C_1 e^{-2(0)} + C_2 e^{-3(0)}$$

$$0 = C_1 e^{0} + C_2 e^{0}$$

$$0 = C_1 + C_2 \Rightarrow \boxed{C_2 = -C_1} \rightarrow (1)$$

$$\frac{dy(x)}{dx} = -2C_1 e^{-2x} - 3C_2 e^{-3x}$$

$\because y'(0) = 15$

$$\Rightarrow 15 = -2C_1 e^{-2(0)} - 3C_2 e^{-3(0)}$$

$$15 = -2C_1 - 3C_2$$

$\because C_2 = -C_1$

$$\Rightarrow 15 = -2C_1 - 3(-C_1)$$

$$= -2C_1 + 3C_1$$

$$\boxed{15 = C_1}$$ sub in eq (1)

$$C_2 = -15$$

$$\Rightarrow C_1 = 15, \quad C_2 = -15.$$

$$y(x) = 15e^{-2x} - 15e^{-3x}.$$

(2) $\frac{d^4x}{dt} + 4x = 0$.

Sol:- Given,

$$\frac{d^4x}{dt^4} = m^4x$$

$$\frac{d^4x}{dt^4} - m^4x = 0$$

In operator form,

$$D^4 - m^4 = 0$$

Auxillary Equation is $r^4 - m^4 = 0$

$$(r^2+m^2)(r^2-m^2) = 0$$

$$(r-m)(r+m)(r^2+m^2) = 0$$

$$r = m, r = -m, r^2 = -m^2$$

$$r = \pm im$$

$$x_c = C_1 e^{mt} + C_2 e^{-mt} + C_3(\cos(mt) + C_4 \sin(mt)$$

General Solution is $x_c = x(m)$

$$\therefore x(m) = C_1 e^{mt} + C_2 e^{-mt} + C_3(\cos(mt) + C_4 \sin(mt).$$

③ $(D^2-6D+9)y = 6e^{3x} + 7e^{-2x} - \log 2$

Given operator form,

$$(D^2-6D+9)y = 6e^{3x} + 7e^{-2x} - \log 2$$

$$f(D)y = Q(x)$$

Auxillary Equation is;

$$f(m) = 0$$

$$m^2 - 6m + 9 = 0$$

$$m^2 - 3m - 3m + 9 = 0$$

$$m(m-3) - 3(m-3) = 0$$

$$(m-3)(m-3) = 0$$

$$m-3=0 \quad | \quad m-3=0$$

$$m=3 \quad | \quad m=3$$

$$m = 3, 3$$

General solution is

$$y_c = C_1 e^{3x} + x C_2 e^{3x}$$

$$y_p = \frac{1}{D^2-6D+9}(6e^{3x} + 7e^{-2x} - \log 2)$$

For the term $\frac{1}{(D-3)^2}(6e^{3x})$

$$= 6\frac{x^2}{2!}e^{3x}$$

$$= 3x^2 e^{3x}$$

for the term $\frac{1}{D^2-6D+9}(7e^{-2x})$

$$= \frac{7e^{-2x}}{(-2)^2-6(-2)+9}$$

$$= \frac{7}{25}e^{-2x}$$

For the constant term $-\log 2$

$$= \frac{1}{D^2-6D+9}[\log 2 \cdot e^{0x}]$$

$$= -\log 2 \frac{e^{0x}}{0^2-6(0)+9}$$

$$= -\frac{\log 2}{9}$$

$$y_p = 3x^2 e^{3x} + \frac{7}{25}e^{-2x} - \frac{\log 2}{9}$$

$$y(x) = y_c + y_p$$

$$\therefore y(x) = (C_1 + C_2 x)e^{3x} + 3x^2 e^{3x} + \frac{7}{25}e^{-2x} - \log 2 - \frac{\log 2}{9}$$

(4) $(D^2+5D-6)y = \sin 4x \sin x$

Sol:- Given in operator form,

$$(D^2+5D-6)y = \sin 4x \sin x$$

$$f(D)y = Q(x)$$

Auxillary Equation $f(m)y = 0$

$$m^2+5m-6=0$$

$$m^2+6m-m-6=0$$

$$m(m+6)-1(m+6)=0$$

$$(m+6)(m-1)=0$$

$$m+6=0 \quad | \quad m-1=0$$

$$m=-6 \quad | \quad m=1$$

$$m=-6,1$$

General solution is,

$$y_c = C_1 e^{x} + C_2 e^{-6x}$$

we know,

$$\sin A \sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$$

$$Q(x) = \frac{1}{2}[\cos(4x-x) - \cos(4x+x)]$$

$$= \frac{1}{2}[\cos 3x - \cos 5x]$$

$$y_p = \frac{1}{D^2+5D-6} Q(x)$$

$$= \frac{1}{2}\left[\frac{1}{D^2+5D-6}\cos 3x - \frac{1}{D^2+5D-6}\cos 5x\right]$$

For first term, $D^2 = (-3)^2 = 9$

$$\frac{1}{D^2+5D-6}\cos 3x = \frac{\cos 3x}{-9+5D-6}$$

$$= \frac{1}{5(D-3)}\cos 3x$$

multiply & divide by $(D+3)$

$$\frac{(D+3)}{5(D^2-9)}\cos 3x = \frac{(D+3)}{5(-9-9)}\cos 3x$$

$$= -\frac{1}{90}(D+3)\cos 3x$$

$$= -\frac{1}{90}(D\cos 3x + 3\cos 3x)$$

$$= -\frac{1}{90}(-3\sin 3x + 3\cos 3x)$$

$$= \frac{3}{90}(\sin 3x - \cos 3x)$$

$$= \frac{1}{30}(\sin 3x - \cos 3x)$$

For second term

$$D^2 = (5)^2 = -25$$

$$\frac{1}{D^2+5D-6}\cos 5x = \frac{1}{-25+5D-6}\cos 5x = \frac{1}{5D-31}\cos 5x$$

Multiply & Divide with $(5D+31)$

$$\frac{5D+31}{(5D-31)(5D+31)}\cos 5x = \frac{5D+31}{25D^2-961}\cos 5x$$

$$= \frac{5D+31}{25(-25)-961}\cos 5x$$

$$= \frac{5D+31}{-625-961}\cos 5x$$

$$= \frac{(5D+31)}{-1586}\cos 5x$$

$$= -\frac{1}{1586}(5D\cos 5x + 31\cos 5x)$$

$D = -25$

$$= \frac{1}{1586}[25\sin 5x - 31\cos 5x]$$

$$y_p = \frac{1}{2}\left[\frac{1}{30}(\sin 3x - \cos 3x) - \frac{1}{1586}\{25\sin 5x - 31\cos 5x\}\right]$$

$$= \frac{1}{60}(\sin 3x - \cos 3x) - \frac{1}{3172}(25\sin 5x - 31\cos 5x)$$

General solution is

$$y = y_c + y_p$$

$$y = C_1 e^{0x} + C_2 e^{-6x} + \frac{1}{60}(\sin 3x - \cos 3x) - \frac{1}{3172}(25\sin 5x - 31\cos 5x)$$

(5) $(D^3 + 4D)y = 5 + \sin 2x + 3^x$

Sol:- Given In operator form,

$$(D^3 + 4D)y = 5 + \sin 2x + 3^x$$

Auxiliary Equation is $f(m)y = 0$

$$m^3 + 4m = 0$$

$$m(m^2 + 4) = 0$$

$m = 0$ | $m^2 = -4$

$m = \sqrt{(2i)^2}$

$m = \pm 2i$

$m = 0, \pm 2i$

$$y_c = C_1 e^{\alpha x} + e^{\alpha x}[C_2 \cos(\beta x) + C_3 \sin(\beta x)]$$

$$= C_1 + C_2 \cos(2x) + C_3 \sin(2x)$$

$$Q(x) = 5 + \sin 2x + 3^x.$$

$$y_{p_1} = \frac{1}{D^3+4D}(5) = \frac{1}{D^3+4D}(5e^{0x})$$

$\because$ Denominator is 0 i.e case failure

$$= x\frac{1}{3D^2+4}(5e^{0x})$$

$$y_{p_1} = \frac{5x}{3(0)^2+4} = \frac{5x}{4}$$

$$y_{p_2} = \frac{1}{D(-4)+4D} \sin(2x)$$

$$= \frac{1}{-4D+4D} \sin(2x)$$

$\because$ Denominator is 0 i.e case failure.

$$= x\left(\frac{1}{3D^2+4}[\sin(2x)]\right.$$

$$= x\left(\frac{1}{3(-4)+4}\sin(2x)\right. = -\frac{x\sin 2x}{8}$$

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$$y_{P_3} = \frac{1}{D^3+4D}(3^x) = \frac{1}{D^3+4D}(e^{x\log 3})$$

For e^{ax}, we replace D with a, Here $a = \log 3$

$$y_{P_3} = \frac{1}{(\log 3)^3 + 4(\log 3)}(e^{x\log 3}) = \frac{3^x}{(\log 3)^3 + 4\log 3}$$

$$y_P = y_{P_1} + y_{P_2} + y_{P_3}$$

$$= \frac{5x}{4} + \left(-\frac{x\sin 2x}{8}\right) + \frac{3^x}{(\log 3)^3 + 4\log 3}$$

$$y = y_C + y_P$$

$$= C_1 + C_2\cos(2x) + C_3\sin(2x) + \frac{5x}{4} - \frac{x\sin 2x}{8} + \frac{3^x}{(\log 3)^3 + 4\log 3}$$

velocity

proport

ela

⑥ $\frac{d^2y}{dx^2} + \frac{dy}{dx} = x^2 + 2x + 4$ ①

Sol:- In operator form,

$(D^2 + D)y = x^2 + 2x + 4$

Auxiliary Equation is $f(m) = 0$

$m^2 + m = 0$

$m(m+1) = 0$

$m = 0 \mid m+1 = 0$

$m = -1$

$m = 0, -1$

General solution,

$y_c = C_1 + C_2 e^{-x}$

$y_p = \frac{1}{D^2+D}(x^2+2x+4)$

$= \frac{1}{D(D+1)}(x^2+2x+4)$

$= \left(\frac{1}{D} - \frac{1}{D+1}\right)(x^2+2x+4)$

$= \frac{1}{D}(x^2+2x+4) - \frac{1}{(D+1)}(x^2+2x+4)$

First term

$= \int x^2 + 2x + 4$

$= \frac{x^3}{3} + x^2 + 4x$

second term

$\therefore \frac{1}{D+a} f(x) = e^{-ax} \int e^{ax} f(x)\, dx$

Here, $a = 1$

$= e^{-x} \int e^x (x^2+2x+4)\, dx$

$= e^{-x}[e^x(x^2+4)]$

$= x^2 + 4$

$y_p = \frac{x^3}{3} + x^2 + 4x - x^2 + 4$

$y_p = \frac{x^3}{3} + 4x - 4$

$y = y_c + y_p$

$\therefore y = C_1 + C_2 e^{-x} + \frac{x^3}{3} + 4x.$

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$$(D^2-4D+4)y = e^{2x} + 8x^2 + \frac{1}{x^2}$$ (c)

Given in operator form,

$$(D^2-4D+4)y = e^{2x} + 8x^2$$

$$f(D)y = Q(x)$$

Auxiliary Equation is $f(m)y=0$

$$m^2-4m+4=0$$

$$(m-2)^2=0$$

$$m=2,2$$

$$y_c = (C_1+C_2x)e^{2x}$$

$$y_p = y_{P_1} + y_{P_2}$$

$$y_{P_1} = \frac{1}{(D-2)^2}e^{2x}$$

If $D=2$, $\Rightarrow$ Case failure

$$= \frac{x^2}{2}e^{2x}$$

$$\because \frac{1}{(D-a)^n}e^{ax} = \frac{x^n}{n!}e^{ax}$$

$$y_{P_2} = \frac{8x^2}{(D-2)^2}$$

$$\because \frac{1}{(D-2)^2} = \frac{1}{4}\left(1+\frac{D}{2}+\frac{3D^2}{4}+\cdots\right)$$

$$D(x^2) = 2x$$

$$D^2(x^2) = 2$$

$$\Rightarrow y_{P_2} = \frac{1}{4}\left[8x^2 + \frac{1}{2}(16x) + \frac{3}{4}(16)\right.$$

$$y_{P_2} = \frac{1}{4}\left[8x^2 + 8x + 12\right]$$

$$= 2x^2 + 2x + 3$$

$$y_p = \frac{x^2}{2} e^{2x} + 2x^2 + 2x + 3$$

$$y = y_c + y_p$$

$$= \left(C_1 + C_2 x\right) e^{2x} + \frac{x^2}{2} e^{2x} + 2x^2 + 2x + 3$$

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(8) $(D^2 - 1)y = x \sin 3x + \cos x$.

Sol:- Given in operator form,

$$(D^2 - 1)y = x \sin 3x + \cos x.$$

$$f(D)y = Q(x)$$

Auxiliary Equation is $f(m) = 0$

$$m^2 - 1 = 0$$

$$(m-1)(m+1) = 0$$

$m - 1 = 0$ | $m + 1 = 0$

$m = 1$ | $m = -1$

$m = 1, -1$

$$y_c = C_1 e^{x} + C_2 e^{-x}.$$

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$$y_p = \frac{1}{D^2-1}(x\sin 3x + \cos x)$$

$$y_p = y_{p_1} + y_{p_2}$$

$$y_{p_1} = \frac{1}{D^2-1} x\sin 3x$$

$$= x\frac{1}{D^2-1}\sin 3x - \frac{2D}{(D^2-1)^2}\sin 3x$$

$$\frac{1}{f(D)} xv = x\frac{1}{f(D)}v - \frac{f'(D)}{[f(D)]^2}v$$

$$D^2 = -a^2$$

$$= -3^2$$

$$= -9$$

$$\Rightarrow x\frac{1}{-9-1}\sin 3x = x\frac{1}{-10}\sin 3x = -\frac{x\sin 3x}{10}$$

$$\frac{2D}{(-9-1)^2}\sin 3x = \frac{2D}{(-10)^2}\sin 3x = \frac{2D}{100}\sin 3x$$

$$= \frac{D}{50}\sin 3x$$

$$= \frac{1}{50} 3\cos 3x$$

$$= \frac{\cos 3x}{4}$$

$$\Rightarrow y_{p_1} = -\frac{x\sin 3x}{10} + \frac{3\cos 3x}{100}$$

$$y_{p_2} = \frac{1}{D^2-1} \cos x$$

$$= \frac{x}{2D} \cos x$$

$$= \frac{x}{2} \int \cos x$$

$$= \frac{1}{2} \cos x$$

$$y_p = y_{p_1} + y_{p_2}$$

$$= -\frac{x \sin 3x}{40} + \frac{3 \cos 3x}{200} + \frac{1}{2} \cos x$$

General Solution is $y = y_c + y_p$

$$y = C_1 e^x + C_2 e^{-x} - \frac{x}{40} \sin 3x + \frac{3}{200} \cos 3x - \frac{1}{2} \cos x$$

(9) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 3y = x^2 e^{-3x}$

Sol:- In Operator form,

$$(D^2 + 2D - 3)y = x^2 e^{-3x}$$

$$f(D)y = Q(x)$$

Auxiliary Equation $f(m)y = 0$

$$m^2 + 2m - 3 = 0$$

$$m^2 + 3m - m - 3 = 0$$

$$m(m+3) - 1(m+3) = 0$$

$$(m+3)(m-1) = 0$$

$m+3=0$	$m-1=0$
$m=-3$	$m=1$

$$m = 1, -3$$

$$y_c = C_1 e^{x} + C_2 e^{-3x}$$

$$y_p = \frac{1}{(D-1)(D+3)}(x^2 e^{-3x})$$

$$y_p = \frac{1}{D^2+2D-3}\left(x^2 e^{-3x}\right)$$

$$\frac{1}{f(D)}\left[e^{ax} V(x)\right] = e^{ax}\frac{1}{f(D+a)} V(x)$$

Here, $a = -3$ $\quad V(x) = x^2$

$$y_p = e^{-3x}\frac{1}{(D-3)^2+2(D-3)-3}x^2$$

$$= e^{-3x}\frac{1}{(D^2-6D+9)+(2D-6)-3}x^2$$

$$= e^{-3x}\frac{1}{D^2-4D}x^2 \qquad (01)$$

$$y_p = e^{-3x}\frac{1}{-4D\left(1-\frac{D}{4}\right)}x^2 = -\frac{e^{3x}}{4D}\left(1-\frac{D}{4}\right)^{-1}x^2$$

$$y_p = e^{-3x}\frac{1}{D^2-4D}x^2$$

$$\left(1-\frac{D}{4}\right)^{-1} \triangleq 1+\frac{D}{4}+\frac{D^2}{16}$$

$$y_p = -\frac{e^{-3x}}{4D}\left[1+\frac{D}{4}+\frac{D^2}{16}\right]x^2$$

$$y_p = -\frac{e^{-3x}}{4D}\left[x^2+\frac{1}{4}(2x)+\frac{1}{16}(2)\right]$$

$$y_p = -\frac{e^{-3x}}{4D}\left[x^2+\frac{x}{2}+\frac{1}{8}\right]$$

$\frac{1}{D}$ means integration.

$$y_p = -\frac{e^{-3x}}{4}\int\left(x^2+\frac{x}{2}+\frac{1}{8}\right)dx$$

$$= -\frac{e^{-3x}}{4}\left(\frac{x^3}{3}+\frac{x^2}{4}+\frac{x}{8}\right)$$

$$y = y_c + y_p$$

$$y = C_1e^{-3x} + C_2e^{x} - \frac{e^{-3x}}{4}\left(\frac{x^3}{3}+\frac{x^2}{4}+\frac{x}{8}\right)$$

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(10) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = xe^x \sin x.$

Sol:- Given In operator form,

$$(D^2-2D+1)y = xe^x \sin x.$$

$$f(D)y = Q(x)$$

Auxiliary Equation $f(m) = 0$

$$m^2 - 2Dm + 1 = 0$$

$$m^2 - m - m + 1 = 0$$

$$m(m-1) - 1(m-1) = 0$$

$$(m-1)(m-1) = 0$$

$$m-1=0 \quad | \quad m-1=0$$

$$m=1 \quad | \quad m=1$$

$$m=1,1$$

$$y_c = (C_1 + xC_2)e^x.$$

$$y_p = \frac{1}{D^2-2D+1}(xe^x \sin x)$$

$$\therefore \frac{1}{f(D)}(e^{ax}V) = e^{ax}\frac{1}{f(D+a)}V$$

$$y_p = e^x \frac{1}{(D+1)^2-2(D+1)+1}(x\sin x)$$

Here $a = 1$

Denominator $= (D+1)^2 - 2(D+1) + 1$

$= D^2 + 1 + 2D - 2D - 2 + 1$

$= D^2$

$$y_p = e^x \frac{1}{D^2}(x\sin x)$$

$\frac{1}{D^2}$ means two times integration.

first integral $= \int x\sin x$

$= -x\cos x + \sin x$

second integral $= \int(-x\cos x + \sin x)dx$

$= -x\sin x - \cos x - \cos x$

$= -x\sin x - 2\cos x$

Thus $y_p = e^x(-x\sin x - 2\cos x)$

General solution $y = y_c + y_p$

$\therefore y = (C_1 + xC_2)e^x + e^x(-x\sin x - 2\cos x)$.

(11) $(D^2-4D+4)y = 8x^2e^{2x}\sin 2x$

Sol:- Given D.E is,

$$(D^2-4D+4)y = 8x^2e^{2x}\sin 2x \quad \text{---}$$

$f(D)y = Q(x)$

Auxiliary Equation is $f(m) = 0$

$m^2-4m+4 = 0$

$(m-2)^2 = 0$

$m = 2, 2$

$y_c = (C_1+xC_2)e^{2x}$

$y_p = \frac{1}{D^2-4D+4} 8x^2e^{2x}\sin 2x$

$= 8\frac{1}{D^2-4D+4} e^{2x}(x^2\sin 2x)$

$= 8e^{2x}\frac{1}{(D+2)^2-4(D+2)+4} x^2\sin 2x$

$= 8e^{2x}\frac{1}{D^2+4D+4-4D-8+4} x^2\sin 2x$

$= 8e^{2x}\frac{1}{D^2} x^2\sin 2x$

$= 8e^{2x}\frac{1}{D^2}$ I.P of x^2e^{i2x}

$\because [e^{i2x} = \cos 2x + i\sin 2x]$

$= 8e^{2x}\frac{1}{D^2}$ I.P of x^2e^{2ix}

$= 8e^{2x}\frac{1}{D^2}$ I.P of $e^{2ix}x^2$

$= 8e^{2x}$ I.P of $e^{2ix}\frac{1}{(D+2i)^2}x^2$

$= 8e^{2x}$ I.P of $e^{2ix}\frac{1}{[2i(1+\frac{D}{2i})]^2}$

$= 8e^{2x}$ I.P of $e^{2ix}\frac{1}{4i^2}\left(1+\frac{D}{2i}\right)^{-2}$

$= 8e^{2x}$ I.P of $e^{2ix}\frac{1}{-4}\left[1-2\left(\frac{D}{2i}\right)+3\left(\frac{D^2}{(2i)^2}\right)-4\left(\frac{D^3}{(2i)^3}\right)+\cdots\right]$

$= -2e^{2x}$ I.P of $e^{2ix}\left[1+\frac{D}{i}-\frac{3D^2}{4}+\frac{D^3}{2i}+\cdots\right]$

$= -2e^{2x}$ I.P of $e^{2ix}\left[1-i\frac{D}{2}-\frac{3D^2}{4}+i\frac{D^3}{2}+\cdots\right]$

$= -2e^{2x}$ I.P of $(\cos 2x+i\sin 2x)\left[1-i\frac{D}{2}-\frac{3D^2}{4}+i\frac{D^3}{2}+\cdots\right]$

$= -2e^{2x}$ I.P of $\left[\cos 2x - i\frac{D}{2}\cos 2x - \frac{3D^2}{4}\cos 2x + i\frac{D^3}{2}\cos 2x + \cdots + i\sin 2x - i^2\frac{D}{2}\sin 2x - \frac{3D^2}{4}\sin 2x + i^2\frac{D^3}{2}\sin 2x + \cdots\right]$

$= -2e^{2x}$ I.P of $\left[\cos 2x - i\frac{D}{2}\cos 2x - \frac{3D^2}{4}\cos 2x + i\frac{D^3}{2}\cos 2x + \cdots + i\sin 2x + \frac{D}{2}\sin 2x - \frac{3D^2}{4}\sin 2x - i\frac{D^3}{2}\sin 2x + \cdots\right]$

$= -2e^{2x}\left[-\frac{1}{2}\cos 2x + \frac{1}{2}\sin 2x\right]$

$= e^{2x}(\sin 2x - \cos 2x)$

$y = y_c + y_p$

$y = (C_1+xC_2)e^{2x} + e^{2x}(\sin 2x - \cos 2x)$

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$$= -2e^{2x} \text{ I.P of } e^{2ix}\left[x^2 - \frac{1}{i}Dx^2 + \frac{3}{4i^2}D^2x^2\right]$$

$$= -2e^{2x} \text{ I.P of } e^{2ix}\left[x^2 - \frac{1}{i}x2x + \frac{3}{4}x2\right]$$

$$= -2e^{2x} \text{ I.P of } e^{2ix}\left[x^2 - \frac{2x}{i} + \frac{3}{2}\right]$$

$$= -2e^{2x} \text{ I.P of } e^{2ix}\left[x^2 - \frac{1}{i}x\frac{1}{i}x2x - \frac{3}{2}\right]$$

$$= -2e^{2x} \text{ I.P of } e^{2ix}\left[x^2 + 1x2x - \frac{3}{2}\right]$$

$$= -2e^{2x} \text{ I.P of } e^{2ix}\left[\frac{2x^2+4x-3}{2}\right]$$

$$= -e^{2x} \text{ I.P } (\cos 2x + i\sin 2x)(2x^2+4x-3)$$

$$y_p = -e^{2x}(4x\cos 2x + 2x^2\sin 2x - 3\sin 2x)$$

General solution is

$$y(x) = y_c + y_p$$

$$y(x) = (c_1 + xc_2)e^{2x} + e^{2x}(4x\cos 2x + 2x^2\sin 2x - 3\sin 2x)$$

(12) $y'' - 6y' + 9y = \frac{e^{3x}}{x^2}$ by the method of variation of parameters.

Sol: In operator form,

$$(D^2 - 6D + 9)y = \frac{e^{3x}}{x^2}$$

$$f(D)\,y = Q(x)$$

Auxillary Equation is $f(m) = 0$

$$m^2 - 6m + 9 = 0$$

$$m^2 - 3m - 3m + 9 = 0$$

$$m(m-3) - 3(m-3) = 0$$

$$(m-3)(m-3) = 0$$

$$m = 3, 3$$

$$y_c = C_1 e^{3x} + xC_2 e^{3x}$$

Let, $y_1 = e^{3x}$, $y_2 = xe^{3x}$

$$y_1' = e^{3x}, \quad y_2' = 3xe^{3x} + e^{3x}$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{3x} & xe^{3x} \\ e^{3x} & 3xe^{3x} + e^{3x} \end{vmatrix}$$

$$= e^{3x}(3xe^{3x} + e^{3x}) - e^{3x}(xe^{3x}$$

$$W = e^{3x}\left(e^{3x} + 3x e^{3x}\right) - 3x e^{3x} e^{3x}$$

$$= e^{6x} + 3x e^{6x} - 3x e^{6x}$$

$$W = e^{6x}$$

$$y_p = -y_1 \int \frac{y_2 Q(x)}{W} dx + y_2 \int \frac{y_1 Q(x)}{W} dx + C$$

$$= -e^{3x} \int \frac{x e^{3x} \cdot e^{3x}}{x^2 e^{6x}} dx + x e^{3x} \int \frac{e^{3x} e^{3x}}{x^2 e^{6x}} + C$$

$$= -e^{3x} \int \frac{1}{x} dx + x e^{3x} \int \frac{1}{x^2} + C$$

$$= -e^{3x} \log x + x e^{3x}\left(-\frac{1}{x}\right) + C$$

$$y_p = -e^{3x} \log x - e^{3x} + C$$

$$y_p = -e^{3x}(\log x + 1) + C$$

$$y(x) = y_c + y_p$$

$$y(x) = (C_1 + xC_2)e^{3x} - e^{3x}(\log x + 1) + C$$

(13) $\frac{d^2y}{dx^2} + a^2y = \tan ax$ by the method of variation of parameters.

Sol:- In operator form,

$$(D^2 + a^2)y = \tan ax$$

Auxiliary Equation $= f(m) = 0$

$$m^2 + a^2 = 0$$

$$m^2 = a^2 i^2$$

$$m = \pm ai$$

$$y_c = e^{0x}(C_1 \cos ax + C_2 \sin ax)$$

$$y_c = C_1 \cos ax + C_2 \sin ax$$

Let,

$y_1 = \cos ax \quad y_2 = \sin ax$

$y_1' = -a \sin ax \quad y_2' = a \cos ax$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

$$= \begin{vmatrix} \cos ax & \sin ax \\ -a \sin ax & a \cos ax \end{vmatrix}$$

$$= a \cos^2 ax + a \sin^2 ax \qquad (W = a)$$

$$= a$$

$Q(x) = \tan ax.$

$$y_p = -y_1 \int \frac{y_2 Q(x)}{W} dx + y_2 \int \frac{y_1 Q(x)}{W} dx + C \quad (1)$$

$$= -\cos ax \int \frac{\sin ax \tan ax}{a} + \sin ax \int \frac{\cos ax \tan ax}{a} dx$$

$$= -\cos ax \int \sin ax \left(\frac{\sin ax}{\cos ax}\right) dx$$

$$+ \sin ax \int \cos ax \left(\frac{\sin ax}{\cos ax}\right) \frac{1}{a}$$

$$= -\cos ax \int \frac{\sin^2 ax}{a \cos ax} dx + \sin ax \int \frac{\sin ax}{a} dx + C$$

$$= -\cos ax \int \frac{1 - \cos^2 ax}{a \cos ax} dx + \sin ax \int \frac{\sin ax}{a} dx + C$$

$$= -\cos ax \int \left(\frac{1}{a} \sec ax - \frac{1}{a} \cos ax\right)$$

$$+ \sin ax \left(\frac{1}{a}\right) \left(-\frac{\cos ax}{a}\right) dx + C$$

$$= -\frac{\cos ax}{a} \left[\log |\tan ax + \sec x| - \sin ax\right]$$

$$- \frac{1}{a^2} \sin ax \cos ax + C$$

$$y_p = -\frac{\cos ax}{a^2} \left[\log |\tan ax + \sec ax|\right]$$

$$y(x) = y_c + y_p$$

$$= C_1 \cos(ax) + C_2 \sin(ax) - \frac{(\cos(ax))}{2} \log |\sec(ax) + \tan(ax)|$$

(17) A Condenser of Capacity 'C' discharged through an inductance 'L' and Resistance 'R' in Series and the charge 'q' at anytime 't' satisfies, $L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{C}$. If $L = 0.25\ H$, $R = 250\ \Omega$, $C = 2\times10^{-6}$ farads then find the charge 'q' at anytime $t>0$ Given that at initially $q = 0.002$ Coulombs and Current $i = \frac{dq}{dt} = 0$ in the Circuit.

Sol:- The differential Equation associated with L·C·R Circuit is,

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{C} = 0$$

$$\Rightarrow \frac{d^2q}{dt^2} + \frac{R}{L}\frac{dq}{dt} + \frac{1}{LC}q = 0 \quad \rightarrow (1)$$

Given that,

$$L = 0.25\ H$$

$$R = 250\ \Omega$$

$$C = 2\times10^{-6}$$

then Eq(1) becomes,

$$\frac{d^2q}{dt^2} + \frac{250000}{6.25}\frac{dq}{dt} + \frac{1}{6.25\times 2\times 10^{-6}}q = 0$$

$$\frac{d^2q}{dt^2} + \frac{25000}{25}\frac{dq}{dt} + \frac{10^6}{0.5}q = 0$$

$$\frac{d^2q}{dt^2} + 1000\frac{dq}{dt} + \frac{10^6}{\left(\frac{1}{2}\right)}q = 0$$

$$\frac{d^2q}{dt^2} + 1000\frac{dq}{dt} + \left(2\times 10^6\right)q = 0$$

The operator form of above DE is

$$\left(D^2 + 1000D + 2\times 10^6\right)q = 0$$

Auxiliary Equation is f(m) = 0

$$m^2 + 1000m + 2\times 10^6 = 0$$

$$m = \frac{-1000 \pm \sqrt{(10^3)^2 - 4\cdot 1\cdot 2\times 10^6}}{2}$$

$$= \frac{-1000 \pm \sqrt{(10^3)^2(1-8)}}{2}$$

$$= \frac{-1000 \pm 10^3\sqrt{7}}{2}$$

$$= \frac{-1000 \pm 10^3\sqrt{7}\,i}{2}$$

$$= \frac{-1000}{2} \pm i\frac{1000}{2}\sqrt{7}$$

$$= -500 \pm i\,500\sqrt{7}$$

$$m = -500 \pm i\,1323$$

$$q_c = e^{-500t}\left(C_1 \cos 1323t + C_2 \sin 1323t\right)$$

$$q(t) = q_c$$

$$\therefore q(t) = e^{-500t}\left(C_1 \cos 1323t + C_2 \sin 1323t\right)$$

$$\frac{dq}{dt} = -500\,e^{-500t}\left(C_1 \cos 1323t + C_2 \sin 1323t\right)$$

$$+ e^{-500t}\left(-1323\,C_1 \sin 1323t + 1323\,C_2 \cos 1323t\right) \rightarrow (2)$$

$q_0 = 0.002$ at $t=0$

① becomes

$$C_1 = q(0) = 0.002 \qquad \therefore C_1 = 0.002$$

At $t=0$, $\frac{dq}{dt} = 0$, ② becomes

$$-500\, e^{-500(0)}(c_1 + 0) + 1323\, c_2 = 0$$

$$-500\,(0.002) + 1323\, c_2 = 0$$

$$1323\, c_2 = 1 \times \frac{2}{10}$$

$$c_2 = \frac{1}{1323}$$

$$c_2 = 0.00075$$

Sub c_1, c_2 in ①

$$\therefore q(t) = e^{-500t}(0.002\cos 1323t + 0.00075 \sin 1323t)$$

which is the charge in LCR circuit at anytime $t \geq 0$.